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A kindergarten teacher's revealed knowledge in orchestration of mathematical activities

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The aim of this study is to employ the Knowledge Quartet, proposed by Rowland, Huckstep, and Thwaites (2005), in order to characterise the kindergarten teacher's competence when orchestrating researcher designed mathematical activities for 5-year-olds. We are drawing on design research as a methodology in which design principles such as playful learning and inquiry are implemented in the activities. Our analyses show that knowledge-in-action and knowledge-in-interaction are revealed in the orchestration. Hence, the Knowledge Quartet is applicable in the kindergarten context – however, with some modifications due to kindergarten particularities.

Keywords: Design research, kindergarten, Knowledge Quartet, mathematics, orchestration.

Introduction

For about a decade, mathematics has been part of the Norwegian curriculum for kindergarten, a curriculum situated within a social pedagogy tradition. In a kindergarten setting in Norway, a kindergarten teacher (KT) is supposed to empower the children's mathematical explorations. However, the curriculum is not explicit as regards how KTs are to facilitate the children's mathematical explorations. In this study we characterise one KT's orchestration of mathematical activities for 5-year-olds in terms of knowledge employed by the KT. The following research questions are formulated for the study:

> In what ways may the Knowledge Quartet be employed in order to characterise the kindergarten teacher's competence?
>
> What knowledge-in-action and knowledge-in-interaction does a kindergarten teacher reveal in her orchestration of a mathematical activity on geometrical shapes?

The activities were designed by researchers in mathematics education (the authors of this paper among others), as part of an ongoing research and development project called "The Agder project"[i], in order for the participating children to engage with mathematical concepts and ideas.

In this study we use the metaphor of orchestration. By this metaphor we mean that it is the KT who is in charge of and leads the mathematical activity. She has to plan, think forward, act in the moment, follow up children's questions and comments, adapt questions for each of the children, etc. – an important and by no means easy role to fulfil. Our admiration of this complexity is thus considerable when we set out to design the mathematical activities and study how the KTs orchestrate the activities.

Theoretical lens: The Knowledge Quartet

In order to analyse in depth how the Knowledge Quartet may be employed in order to characterise the kindergarten teacher's competence and the revealed knowledge-in-action and knowledge-in-interaction on behalf of the KT, we draw heavily on the Knowledge Quartet coined by Rowland,

Huckstep and Thwaites (2005). The Knowledge Quartet was developed by Rowland et al. (2005) as a theoretical construct drawing on the profound and substantial work of Shulman (1986). Rowland and colleagues drew on videotapes from mathematics classroom lessons where pre-service teachers utilised their mathematical and pedagogical knowledge in their teaching. This quartet is used as a theoretical lens through which we analyse video data of one KT´s orchestration of one of the developed activities. We limit ourselves to one KT taking into account that "the quartet is comprehensive as a tool for thinking about the ways that subject knowledge comes into play in the classroom" (Rowland, Huckstep, & Thwaites, 2003, p. 97).

The Knowledge Quartet differs from the framework of Ball, Thames and Phelps (2008), mathematical knowledge for teaching (MKT), in that the former focuses on situations in the mathematics classroom through which the teacher's mathematics-related knowledge may be observed. Ball et al.'s (2008) framework describes different kinds of mathematics teachers' knowledge. Mosvold, Bjuland, Fauskanger, and Jacobsen (2011) used the framework of Ball et al. (2008) to analyse mathematics teaching at the kindergarten level. Mosvold et al. (2011) found that the MKT framework needs to be adjusted to the kindergarten context in order to be appropriately used. This is because (1) the work situation of orchestrating mathematical activities for a Norwegian kindergarten teacher is very different from the teaching situation of a U.S. mathematics school teacher; and (2) the tasks of teaching in the kindergarten setting are significantly different from that of the school setting. It is pedagogical activities which is the basis for learning activities in Norwegian kindergartens, not mathematical activities as such. These differences have been emphasised by Erfjord, Hundeland, and Carlsen (2012), through the lens of the didactic triangle.

In adopting a grounded approach to the analyses of data, Rowland et al. (2005) identified what they labelled the Knowledge Quartet, four dimensions along which mathematics teachers' "mathematics-related knowledge" (p. 255) may be analysed. These four dimensions are termed *foundation*, *transformation*, *connection* and *contingency*. Foundation encompasses the knowledge background of the KT; transformation and connection encompass how and to what extent knowledge is revealed in action as the KT implements and orchestrates the activity, hence dimensions described as knowledge-in-action. Contingency encompasses the KT's knowledge as this unfolds in interaction with the children, hence knowledge-in-interaction.

Foundation is a dimension of *propositional knowledge* used, adapted to our case, to address the KT's mathematical knowledge, the KT's knowledge of mathematics education, and the KT's view upon the purpose of mathematics education and how children learn mathematics. Analytically, we use this dimension to characterise both the mathematics and the didactical insights revealed by the KT in her orchestration of the activity.

Transformation is a dimension of *knowledge-in-action* which addresses the KT's choices of representations, demonstrations, and use of examples in her orchestration of the activity. The dimension focuses on the KT's ability to transform the mathematics "in ways designed to enable students to learn it" (Rowland et al., 2005, p. 265). Analytically, we use this dimension to describe the KT's orchestration of the activity on characterising two-dimensional geometrical shapes.

Connection is a dimension of *knowledge-in-action* as well, addressing to what extent the KT draws connections between various mathematics concepts and connections between various mathematics

procedures, alternative meanings for these concepts and different ways of carrying out procedures. Analytically, we use this dimension to characterise how the KT makes connections between the geometrical shapes and their features, as well as various ways of deciding what shape is what.

Contingency is a dimension of *knowledge-in-interaction*, a dimension that addresses how the KT interacts with the children through appropriately responding to children's contributions, to what extent she takes advantage of learning opportunities that emerge, and to what extent she makes the activity her own and deviates from the goals and foci of the activity. Analytically, we use this dimension to characterise how the KT responds to the children's ideas, the questions and comments she uses, whether she draws the children's attention towards particular mathematical ideas, and whether she makes the geometry activity her own and are unbounded by its original foci.

Design principles and context

In our design of the mathematical activities we drew on two main principles, *playful learning* and *inquiry approach* to the teaching and learning of mathematics. The design principle of playful learning emphasises that for children play and learning is one and the same thing. Playful learning encompasses both free play (child-initiated and child-directed play) and guided play (adult-initiated and child-directed play), where guided play is the principle used here. In guided play the KT orchestrates and literally guides the play in an adequate direction in order to reach pre-formulated aims for the children's play and to nurture the children's interest, curiosity, engagement, and (mathematical) sense-making (Weisberg, Kittredge, Hirsh-Pasek, Golinkoff, & Klahr, 2015). The design principle of adopting an inquiry approach to the teaching and learning of mathematics stems from Jaworski's (2005) inquiry as "a way of being in practice" (p. 103). That is, the KT and the children collaborate in order to achieve meaningful answers to prompts and questions, the children's curiosity is taken advantage of, and the children are guided into mathematical inquiries through the use of questions, being curious and excited about mathematical issues.

The KT that takes part in our study is a participant in the Agder Project, and by that also a participant in a professional development program. The professional development program within this project was based on four workshops of two days duration, focusing on Number, Measurement, Geometry and Statistics, combinatory and probability. In the third workshop, the KTs participated in lectures and group discussions on Geometry. Our data collection took place during the period when the KTs tried out two geometry activities. The observed KT is in her forties, educated for three years (180 ECTS) at university level, and is a well experienced kindergarten teacher.

Context

We as researchers are interested in scrutinising the KTs' processes of orchestrating our designed mathematical activities. Moreover, the KTs have been participants in a professional development program where we have contributed with lectures and feedback on previous orchestrations of mathematical activities. We study the case of Wilma and videotaped her orchestration of a geometry activity. Wilma had received a written instruction note for the geometry activity a couple of weeks in advance of our data collection.

Analysis and results

In order to employ the Knowledge Quartet (KQ) on empirical data, one has to go into the code level from which the four dimensions are extracted (Rowland, 2016, personal communication). Our analytical process started with a first phase of collectively reflecting on our collected data in total, consisting of two days with data collection in each of four different kindergartens. In doing that, we had the KQ in mind, seeking to use KQ terminology and codes to analyse our data. The second phase consisted of us collectively looking at video excerpts from the different kindergartens, resulting in our choice of the case of Wilma. The reason for choosing one of the sessions of Wilma was that this session, from our collective looking at the video, turned out to be the most promising in addressing our research questions. The KT observably revealed her knowledge in action and interaction, the children contributed with oral statements and questions, and interaction amongst the children was taking place. The third analytical phase consisted of transcribing the video from Wilma's session. The fourth phase consisted of us conducting collective in-depth analyses of the natural talk-in-interaction adopting the codes developed in the Knowledge Quartet.

The activities were presented in written form by us with the three headings *Equipment*, *Intention,* and *Implementation.* The particular activity we consider here, is the first part of an activity focusing on two-dimensional geometrical shapes. Under *Equipment* we wrote: Geometrical shapes: triangles, squares, rectangles, circles, trapezium, rhombus, while for *Intention* we wrote*:* The children are supposed to get experience in recognizing properties of different two-dimensional shapes. Furthermore, the children are supposed to practice mathematical argumentation with respect to features of the various shapes. As regards *Implementation,* we wrote: Let the children investigate the shapes and their characteristics. Let the children discover the shapes' differences.

Excerpt 1: Foundation

The following excerpts are taken from the initial phase of Wilma's orchestration of the geometry activity. Wilma shows and shakes a box containing two-dimensional paper shapes. One of the children responds that "It's shapes". Upon this, Wilma continues their conversation. The analytical contributory codes are: "awareness of purpose; identifying errors; overt subject knowledge; theoretical underpinning of pedagogy; use of terminology; use of textbook; reliance on procedures" (Rowland et al., 2005, p. 265).

Wilma:	Yes, that's correct. And with mathematical terminology we call them geometrical shapes. Are you able to pronounce that?
Sam:	I think it is cookies (Smiles as he says it).
Wilma:	John, are you able to pronounce that? Geometrical shapes?
John:	Geometrical shapes. (Several children simultaneously say "Geometrical shapes")
Wilma:	Yes, that is what they are called with mathematical terminology. Inside this box there are several of such shapes (She opens the box and shows it to all the children so that they may look inside the box).
John:	It looks like a puzzle.
Wilma:	Yes, it looks like a puzzle. That's true.

Sam: Yes. Are we going to puzzle with them?

Wilma: At least we are going to work with them, yes we are.

Ken: Can you pour them out?

Wilma: I was thinking of pouring them out. Then I want you to take a look at them. Currently, there are quite a few shapes and some of them are almost identical. Now you may take a look at them. (She pours the shapes out on the table; the children take some shapes each and say "that is small" and "a triangle".)

In this excerpt we argue that Wilma's orchestration is characterised by her foundation, both with respect to mathematics insights and didactical insights. The mathematical insights are revealed through Wilma's emphasis on the mathematical terminology through the twice expressed term "geometrical shapes". Furthermore, her mathematical insights are revealed through her choice of shapes that are congruent, shapes that are similar, and the variety of shapes included (various triangles, various quadrilaterals, circles of various sizes, ellipses, hexagons and octagons). The written material made by us suggested "triangles, squares, rectangles, circles, trapezium, rhombus" as shapes to work on in the activity while Wilma introduced many more shapes and terminology. Thus, we argue that the foundation here is Wilma's and not just her use of external provided foundation from us in the written form.

The didactical insights of Wilma are revealed through her way of establishing interest and curiosity among the children by shaking the box. The children's interest and curiosity about the shapes are nurtured further by her showing of the various shapes in the box. Wilma confirms that the shapes look like pieces of a puzzle and by that she makes a link between these shapes and the apparently well-known activity of puzzling. By establishing that link, Wilma also communicates a playful way of engaging with the shapes. Finally, in this excerpt, Wilma's didactical insights are revealed when she asks the children to inquire into the various shapes. By orchestrating the activity as playful, Wilma signals that she appreciates inquiry as a tool in order to learn mathematics. Additionally, the children's interest, curiosity, engagement, and mathematical sense-making are nurtured.

Excerpt 2: Transformation and Connection

Approximately ten minutes later, the following dialogue occurred, exemplifying the dimensions of transformation and connection. Relative to transformation, the analytical codes are: "choice of representation; teacher demonstration; choice of examples" (Rowland et al., 2005, p. 265). Relative to connection, the analytical codes are: "making connections between procedures; making connections between concepts; anticipation of complexity" (Rowland et al., 2005, p. 265). Some of the children had picked up quadrilaterals that they found interesting since they did not know their names and they were different from rectangles and squares. Then the KT said:

Wilma: Do you know what? These two quadrilaterals actually have other names with mathematical terminology. They have four edges (She counts "one, two, three, four" aloud while simultaneously pointing at the edges).

Susie: But what are they called then?

Wilma: That one is called a rhombus (she points at the rhombus while speaking).

Susie: Rhombus.

Wilma: Rhombus. And that one, do you notice that two and two edges are equal (she points at the parallelogram she shows). That edge and that edge (slides her finger along the two opposite, parallel edges), are equal, and that edge and that edge are equal (slides her finger along the two other, opposite, parallel edges). Its name is actually a parallelogram.

Sam: A paragram?

Wilma: Yes, a parallel o gram.

In this excerpt we see that Wilma carefully introduces two new shapes for the children. She gets everyone's attention by holding up and showing one shape at a time. She emphasises that both shapes are conceptually associated with quadrilaterals, by overtly counting the edges. At the same time she makes it obvious that the two shapes are particular kinds of quadrilaterals. This is pinpointed by saying that they have "other names", implicitly distinguishing these shapes from the familiar quadrilaterals square and rectangle.

Wilma's orchestration in this excerpt exemplifies transformation in that she uses one example of each of the two new shapes. She points at each of them in accordance with what she is saying, and she demonstrates how to classify geometrical shapes by counting their edges. Her orchestration also exemplifies contingency due to the deviations from the agenda set by the researchers in the designed activity.

The dimension of connection is also exemplified in this excerpt as Wilma draws the children's attention towards the particular features of one of the shapes. Wilma carefully focuses at the two pairs of edges that are parallel, one pair at a time, by sliding her index finger along the edges. She neither uses the mathematical concept of parallel nor equal length at this occasion. She only uses the feature "equal". The feature of parallelism is thus only implicitly focused. However, by sliding her index finger along the two edges, her gesture signals that they are of equal length.

In the end of the dialogue we also notice that Wilma is particularly focused at offering opportunities for the children to learn the name of the new shape. On two occasions, Wilma uses the name "parallelogram" for the shape. Both times she puts emphasis on the expression "parallelogram". We see that Sam tries to pronounce the name, but only partly succeeds. Wilma then slows down the pace in her pronunciation of the word, in order for Sam, and the other children, to pay attention to the new word. Hence, we observe that Wilma is eager to name new objects mathematically correct. Naming is an important element in these children's mathematical learning process. However, both in this example and in other examples the naming of shapes comes at a late stage, after the children have presented their sorting of shapes and the KT has orchestrated a discussion of the properties of the shapes. We consider this as evidence of her taking an inquiry approach in her orchestration.

Excerpt 3: Contingency

To exemplify this dimension of the KQ in Wilma's orchestration, we include examples of single moves which illustrate how she responded to the children's ideas, the questions and comments she used to make the children pay attention to various mathematical ideas. Even though she occasionally addressed her questions to one particular child, the other children still paid attention.

The analytical codes used were: “responding to children’s ideas; use of opportunities; deviation from agenda” (Rowland et al., 2005, p. 266).

Wilma drew the children’s attention towards the mathematical concept of sorting. Sam said: “Can we sort them?” (19). Then, some moves, but only a few seconds later, Wilma said: “Sam, what does it mean to sort?” (27). Wilma also drew the children’s attention towards mathematical features of the various shapes. One example was found when Sam said: “Yes, but these are small (points at the short edges of the rectangle). These two are equally long” (144). A few seconds later, Wilma responded to this utterance and asked all of the five children: “Does anybody know what the shape is called when two edges are quite long and two edges are shorter?” (147). With this question, she seeks to establish interest and curiosity, and thus to nurture playfulness in her orchestration.

Wilma likewise focused their communication around the similarities between the shapes. Susie talks about the fact that two shapes, two congruent triangles, may be joined in order to make a rectangle. She says: “Jack’s shapes have such…, but Ken’s do not have such when he puts them together” (196). Upon this statement of observation, Wilma responds immediately: “What happens when you put them together?” (197). Furthermore, Wilma addressed the children’s mathematical reasoning through questions like: “How did you figure out that one (points at one of the hexagons)?” (72); and “Do you want to tell the other children?” (227). These examples show that Wilma adopts inquiry as a way of being (cf. Jaworski, 2005) and responds to the children’s ideas and uses the opportunities that unveil in their interaction.

Discussion

In this study we have considered the following questions: In what ways may the Knowledge Quartet be employed in order to characterise the kindergarten teacher’s competence? and What knowledge-in-action and knowledge-in-interaction does a kindergarten teacher reveal in her orchestration of a mathematical activity on geometrical shapes? From what is possible to include in this relatively short paper, we are able to discern instances of all four knowledge dimensions through which her orchestration of the mathematical activity is informed. As such, the Knowledge Quartet (Rowland et al., 2005) has proven to be analytically useful when seeking to wrap up how a KT is able to orchestrate researcher-designed mathematical activities. At the same time, we observe that the KQ unfolds slightly different in the kindergarten context than in the school context, the context from which the KQ was originally elaborated and developed. In the Norwegian kindergarten context, it is highly rare, even inappropriate to orchestrate mathematical activities through long introductions and demonstrations. Concerning the issue of time, we see that the periods where the KT has the word are quite short, often only 5-10 seconds. Furthermore, in the kindergarten context it is unusual to give children extensive time to inquire into the mathematics without KT interference. This is however a usual case in the school context. KQ contributory codes, such as use of textbook, reliance on procedures, teacher demonstration, and making connections between procedures, are partially inappropriate and inapplicable in the kindergarten context. Thus, our modification of the KQ materialised as not taking these codes into consideration.

It is challenging to separate the dimensions from each other in the dialogues. In the dialogues the dimensions are intertwined, where one move may simultaneously exemplify several dimensions. The excerpts above are thus not mutually exclusive when it comes to the four dimensions of the

KQ. For the purpose of this paper, however, the excerpts are deliberately chosen to illustrate how the KT reveals her knowledge-in-action and knowledge-in-interaction with respect to this particular mathematical activity.

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